PRIVACY-PRESERVING DATA ANALYSIS

Lecture 2 - Differential privacy

Static study notes. The interactive version, with live demos and settings, is on the website.

CHAPTER 1 OF 11

From attacks to a definition

Lecture 1 showed that answering questions with noise can still leak a whole secret column. The lesson was a **threshold**: the noise must be around √n.

That's a rule of thumb, not a guarantee. We want a definition that says exactly what an attacker is **not** able to learn about one person, no matter how clever they are or what other data they have.

That definition is **differential privacy**. To work with it we need two tools from probability: **expectation** and **variance**.

WATCH FOR

- The route panel: recap, definition, proof, Laplace noise, composition, playground.
- Each chapter runs a demo by itself. Pause, replay or skip any time.

Lecture 1 ended with a problem

Noisy answers can still leak a whole secret column. We saw two attacks, and we saw that the noise has to be around √n before they fail.

But “noise of size E” is a *recipe*, not a promise. It doesn't say what an attacker can learn about **one person**.

Lecture 2 gives the promise

Differential privacy says: the output must look almost the same whether or not any one person's row is in the table. “Almost” is measured by a number, **ε**.

To prove things we need a little probability first: expectation and variance.

The route

1. Recap: expectation and variance, with demos you can run.
2. The definition of ε-differential privacy.
3. Proof that randomized response is ε-DP, then how to estimate from it.
4. The Laplace mechanism for numeric queries.
5. Composition: ε adds up in series, not in parallel.
6. The privacy budget against the lecture 1 attacker, and a live playground.

Recap 1: expectation

A **random variable** X is a number decided by chance. A biased coin is the simplest one: X = 1 with probability p, otherwise 0.

Its **expectation** is the average value you'd get over a very long run, weighted by probability:

$$E[X] = \Sigma\ x \cdot P(X = x) = 1 \cdot p + 0 \cdot (1-p) = p$$

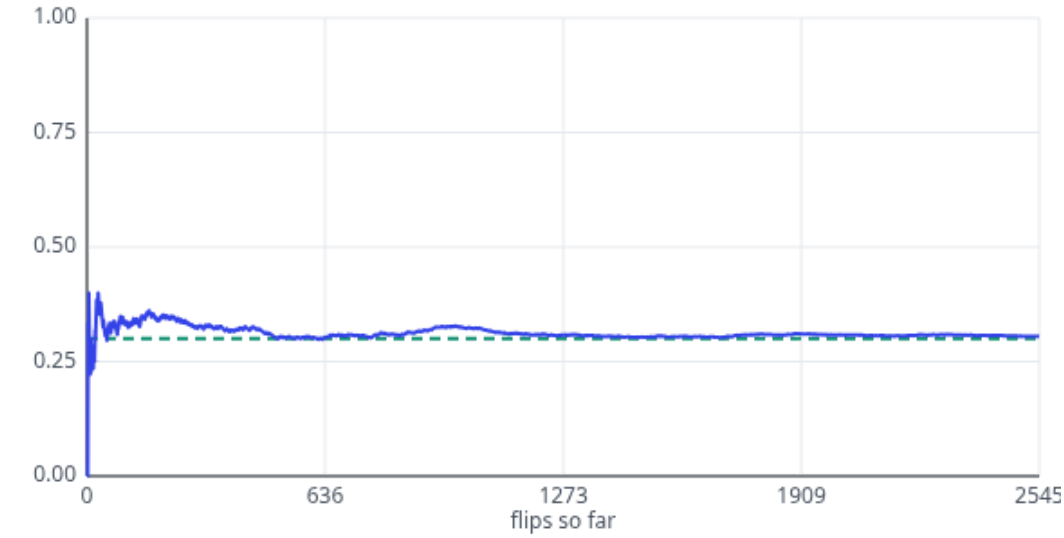
Two rules you'll use all lecture: **E[X+Y] = E[X] + E[Y]** (always, even if X and Y are related) and **E[a·X] = a·E[X]**.

WATCH FOR

- The blue line is the average of the flips so far. It wobbles at first, then settles on the green line, p. This is the *law of large numbers*.

A biased coin

X = 1 with probability p, otherwise 0



FLIPS SO FAR	AVERAGE SO FAR	EXPECTATION E[X]	GAP
2,545	0.305	0.30	0.005
	sum of flips ÷ number of flips	what the average heads towards	shrinks as flips grow

$$E[X] = 1 \cdot P(X=1) + 0 \cdot P(X=0) = 1 \cdot 0.30 + 0 \cdot 0.70 = \mathbf{0.30}$$

$$E[X+Y] = E[X] + E[Y],\quad E[a \cdot X] = a \cdot E[X]$$ (so n flips have expected total $n \cdot p = 30$ for $n = 100$)

last flips: 0 0 1 0 0 0 1 0 1 1 0 0 0 0 0 0 1 0 1 1 0 0 0 0 1 0 1 0 0 1 0 1 0 0 0 0

Recap 2: variance, and where √n comes from

Expectation says where the average sits. **Variance** says how far values typically stray from it:

```
Var(X) = E[(X − E[X])²] = E[X²] − E[X]²
coin: E[X²] = p, so Var(X) = p − p² = p(1−p)
```

Rules: **Var(a·X) = a²·Var(X)**, and for **independent** variables **variances add**: $\mathrm{Var}(X_1+\ldots+X_n) = n\cdot p(1-p)$. The **standard deviation** is the square root, so the spread of a count grows like **√n**, and the spread of an average shrinks like **1/√n**.

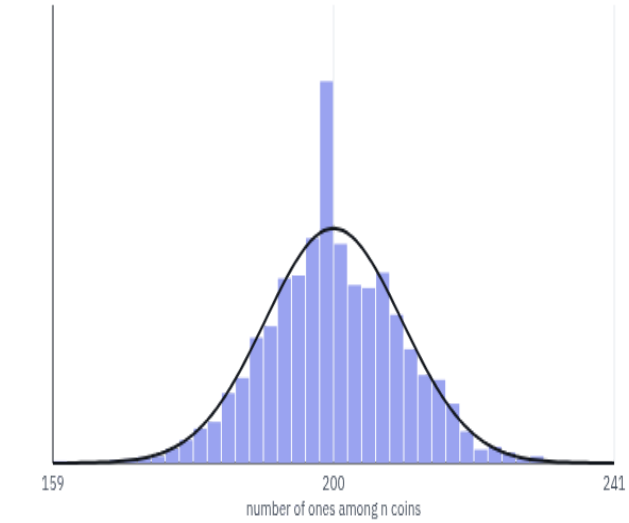
That is the same √n as in lecture 1: noise of size √n is what the data's own randomness looks like.

WATCH FOR

- As n grows the histogram of counts moves right and gets wider, but only like √n (right chart).

Add up n coins

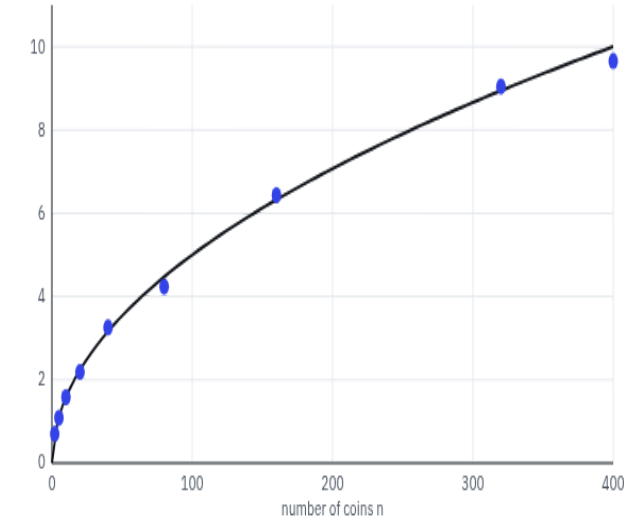
n = 400, p = 0.50



3000 repeated experiments

bell curve with the theory mean and std

Spread of the sum against n



theory √(n·p·(1−p)) measured

MEAN OF THE SUM
200.0
E[sum] = n·p

VARIANCE OF ONE COIN
0.250
Var(X) = E[X²] − E[X]² = p − p² = p(1−p)

STD OF THE SUM
10.00 (measured 10.04)
independent variances add: Var(sum) = n·p(1−p), so std ∝ √n

STD OF THE AVERAGE
0.025
divide by n: std ∝ 1/√n

The definition of differential privacy

Two tables are **neighbours** if they differ in one person's row. A randomized mechanism M is **ε-differentially private** if for all neighbours D, D′ and every set S of possible outputs:

$$\Pr[\, M(D) \in S \,] \le e^{\varepsilon} \cdot \Pr[\, M(D') \in S \,]$$

In words: adding or removing anyone changes the chance of any outcome by at most a factor e^{ε}. Small ε means close to 1, so the output barely depends on any one person. (For small ε, $e^{\varepsilon} \approx 1 + \varepsilon$.)

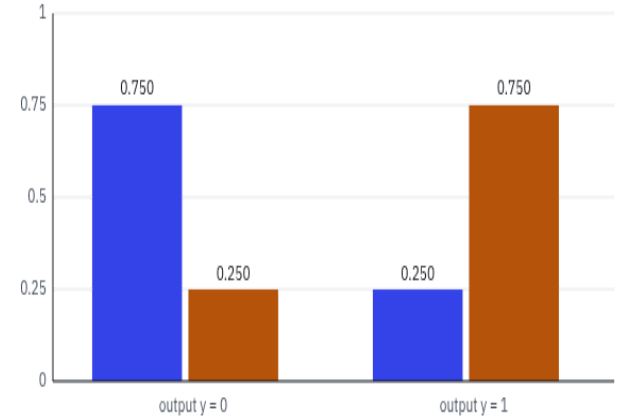
The right panel shows what this means for an attacker: whatever they believed, one answer can only multiply their **odds** by at most e^{ε}.

WATCH FOR

- The bars are the output probabilities for a person with secret 0 (blue) and secret 1 (orange).
- As ε grows, the two bars separate and the attacker learns more.

Randomized response, one person

output y given secret x



secret x = 0 secret x = 1

KEEP THE TRUTH, P	RATIO FOR Y = 0	RATIO FOR Y = 1	LARGEST RATIO
0.750 p = e^ε / (1 + e^ε)	**3.00** P(y=0|x=0) ÷ P(y=0|x=1)	**0.33** P(y=1|x=0) ÷ P(y=1|x=1)	**3.00** = e^ε, never more

What an attacker learns

Bayes, with prior P(x=1)

Before seeing anything: P(sick) = **0.50**.

The person reports y = 1. Bayes' rule gives

$$P(x=1 \mid y=1) = \pi \cdot p \,/\, (\, \pi \cdot p + (1-\pi)(1-p)\,) = \mathbf{0.750}$$

The odds of “sick” went from 1.00 to 3.00: multiplied by exactly **3.00** = e^{ε}. Whatever the attacker believed, one answer can shift the odds by at most a factor e^{ε}.

Proof: randomized response is ε-DP

Randomized response lets people answer a sensitive yes/no question with deniability. Each person tells the truth with probability p and lies with probability 1−p.

Choose $p = e^{\varepsilon}/(1+e^{\varepsilon})$. The **formal proof** (shown below the charts) has four steps: factor the probability of an output over the n independent people, cancel every factor except the one person who differs, see that the remaining ratio is exactly e^{ε} (or $e^{-\varepsilon}$), and sum over outputs to get it for sets. A worked version with your current numbers follows it.

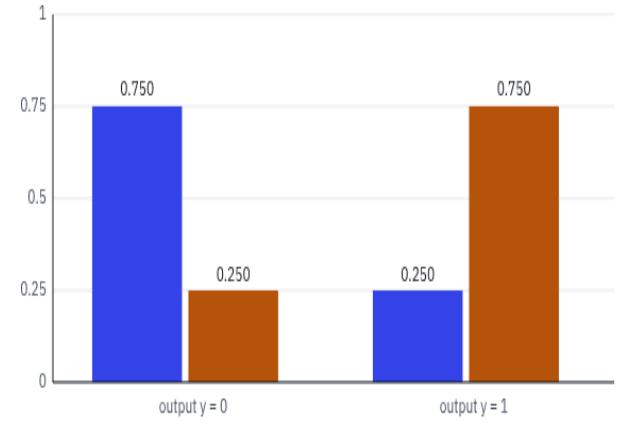
Two-coin version: flip a coin; heads, answer truthfully; tails, flip again and report that coin. Then p = ¾ and ε = ln 3 ≈ 1.1.

WATCH FOR

- The shaded row of the table: the ratios never exceed e^{ε}.
- Drag ε in the settings and the table updates.

Randomized response, one person

output y given secret x



secret x = 0 secret x = 1

KEEP THE TRUTH, P	RATIO FOR Y = 0	RATIO FOR Y = 1	LARGEST RATIO
0.750 p = e^ε / (1 + e^ε)	**3.00** P(y=0|x=0) ÷ P(y=0|x=1)	**0.33** P(y=1|x=0) ÷ P(y=1|x=1)	**3.00** = e^ε, never more

What an attacker learns

Bayes, with prior P(x=1)

Before seeing anything: P(sick) = **0.50**.

The person reports y = 1. Bayes' rule gives

```
P(x=1 | y=1) = π·p / ( π·p + (1−π) (1−p) ) = 0.750
```

The odds of “sick” went from 1.00 to 3.00: multiplied by exactly **3.00** = e^{ε}. Whatever the attacker believed, one answer can shift the odds by at most a factor e^{ε}.

Formal proof that randomized response is ε-DP

SETTING

A database is $D = (x_1, \ldots, x_n) \in \{0,1\}^n$, one secret bit per person. The mechanism $\mathbf{RR_p}$ treats every person independently: it outputs $y_i = x_i$ with probability p, and $y_i = 1 - x_i$ with probability $1 - p$, where $p = e^\varepsilon/(1 + e^\varepsilon)$ (so $\frac{1}{2} \le p < 1$). The released output is $M(D) = (y_1, \ldots, y_n)$. Two databases D, D′ are **neighbours** if they differ in exactly one coordinate j.

THEOREM

For all neighbours D, D′ and every set $S \subseteq \{0,1\}^n$ of outputs: $\mathbf{Pr[M(D) \in S] \le e^\varepsilon \cdot Pr[M(D') \in S]}$.

PROOF

1. One output. The coordinates are independent, so for any $y \in \{0,1\}^n$: $\Pr[M(D) = y] = \prod_i \Pr[y_i \mid x_i]$, where $\Pr[y_i \mid x_i] = p$ if $y_i = x_i$ and $1 - p$ otherwise.

2. The ratio. D and D′ agree outside coordinate j, so the factors for $i \ne j$ are identical and cancel: $\Pr[M(D)=y] / \Pr[M(D')=y] = \Pr[y_j \mid x_j] / \Pr[y_j \mid x'_j]$. The bit is binary and $x_j \ne x'_j$, so exactly one of them equals y_j. The ratio is therefore either $p/(1-p)$ or $(1-p)/p$.

3. The bound. $p/(1-p) = e^\varepsilon$ (by the choice of p), and $(1-p)/p = e^{-\varepsilon} \le e^\varepsilon$ because $\varepsilon \ge 0$. So for every single output y: $\Pr[M(D)=y] \le e^\varepsilon \cdot \Pr[M(D')=y]$.

4. Sets. For any S, $\Pr[M(D)\in S] = \Sigma_{y\in S} \Pr[M(D)=y] \le \Sigma_{y\in S} e^\varepsilon \cdot \Pr[M(D')=y] = e^\varepsilon \cdot \Pr[M(D')\in S]$. ∎

REMARKS

Tight: the ratio in step 2 really equals e^ε for some y, so $\varepsilon = \ln(p/(1-p))$ is exactly the privacy level of RR_p, and no smaller ε works. **Two-coin version:** $p = \frac{3}{4}$ gives $\varepsilon = \ln 3 \approx 1.099$. **Extreme cases:** $p = \frac{1}{2}$ gives $\varepsilon = 0$ (perfect privacy, but the output is pure noise); $p \to 1$ gives $\varepsilon \to \infty$ (no privacy). **Post-processing:** any function of the output, such as the estimator $\hat{\pi}$ in the next chapter, is also ε-DP.

The same proof with your current numbers ε = 1.10, p = 0.750

STEP 1. THE MECHANISM

Each person holds a secret bit x. They report y = x with probability $p = e^\varepsilon/(1+e^\varepsilon) = 0.750$, and $y = 1-x$ with probability $1-p = 0.250$. (Two fair coins: flip one; on heads tell the truth; on tails flip again and report that coin. Then $p = \frac{3}{4}$ and $\varepsilon = \ln 3 = 1.099$.)

STEP 2. PROBABILITY OF EACH OUTPUT

SECRET	P(Y=0)	P(Y=1)
x = 0	0.750 = p	0.250 = 1−p
x = 1	0.250 = 1−p	0.750 = p
ratio P(y|x=0) ÷ P(y|x=1)	**p/(1−p) = 3.004**	**(1−p)/p = 0.333**

STEP 3. COMPARE WITH E^E

$p/(1-p) = e^{\varepsilon}/(1+e^{\varepsilon}) \cdot (1+e^{\varepsilon})/1 = \mathbf{e^{\varepsilon}} = 3.004$. The other ratio is $e^{-\varepsilon} \le e^{\varepsilon}$. So for every output y and in both directions, $\Pr[M(x)=y] \le e^{\varepsilon} \cdot \Pr[M(x')=y]$.

STEP 4. FROM OUTPUTS TO SETS

For any set S of outputs, $\Pr[M(x) \in S] = \Sigma_{y \in S} \Pr[M(x)=y] \le \Sigma_{y \in S} e^{\varepsilon} \cdot \Pr[M(x')=y] = e^{\varepsilon} \cdot \Pr[M(x') \in S]$. That is the definition. ∎

CHAPTER 6 OF 11

Using expectation and variance on the answers

Each person's report y is noisy, but the analyst still wants the true fraction π of sick people. Let q = Pr[y=1] = (1−p) + π(2p−1). The raw average ȳ has E[ȳ] = q, which is **not** π, so correct it:

$$\hat{\pi} = (\bar{y} - (1-p)) \,/\, (2p-1)$$
$$E[\hat{\pi}] = (q - (1-p)) \,/\, (2p-1) = \pi \text{ (unbiased)}$$
$$\mathrm{Var}(\hat{\pi}) = q(1-q) \,/\, \left(n \cdot (2p-1)^2 \right)$$

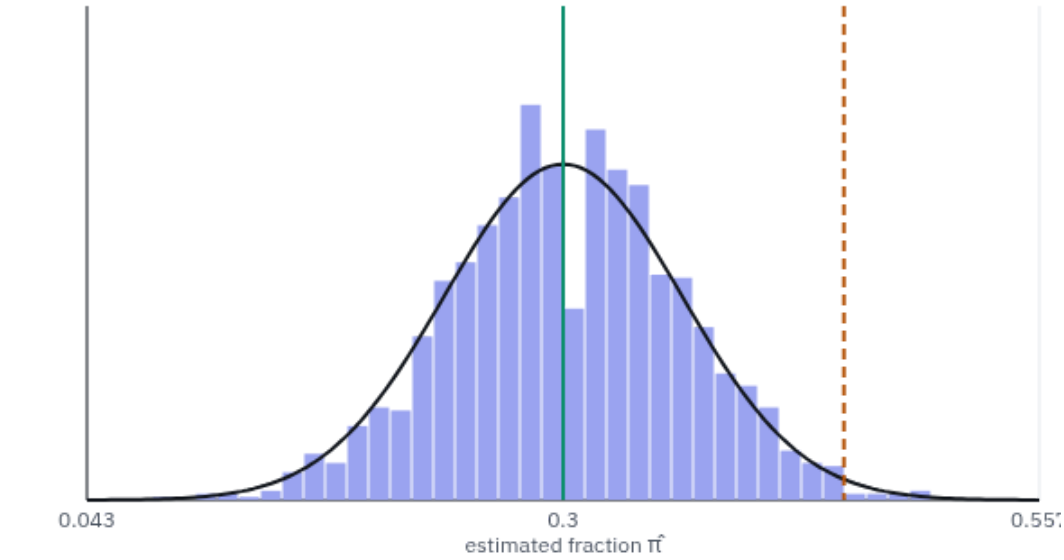
Each y is a coin, so Var(ȳ) = q(1−q)/n by the variance rules. The smaller ε is, the smaller 2p−1 is, so the larger the variance: **privacy costs accuracy**, and the error in people-count is of order √n.

WATCH FOR

- The surveys centre on the green line (unbiased); the dashed orange line (raw average) is off.
- The bell curve from the formulas matches the simulation.

A survey with randomized response

n = 1000, π = 0.30, ε = 0.5



estimate π̂ in 1500 repeated surveys · bell curve from the formulas · true fraction π · raw average of the noisy answers (biased)

AVERAGE OF π̂	RAW AVERAGE OF ANSWERS	STD OF π̂	ERROR IN PEOPLE
0.302	0.452	0.0633	63.3 of 1000
true π = 0.30: unbiased, E[π̂] = π	q = (1−p) + π(2p−1): biased, not π	formula √(q(1−q)/n)/(2p−1) = 0.0642	compare √n = 31.6: grows like √n/ε (lecture 1)

The Laplace mechanism for counts

For numeric queries, add noise directly. A count changes by at most 1 when one person is added or removed: its **sensitivity** is Δ = 1. Release

$$M(D) = \text{count}(D) + \text{Lap}(1/\varepsilon), \text{ density } \propto e^{-\varepsilon \cdot |z - \text{count}|}$$

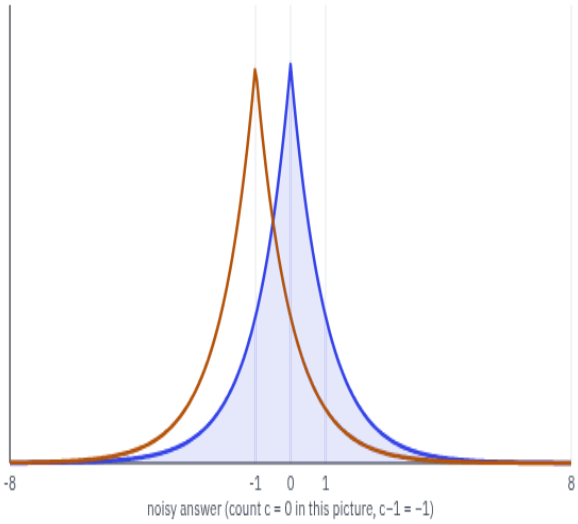
Why it is ε-DP. For any output z, the density ratio between neighbours is $e^{-\varepsilon|z-c|} / e^{-\varepsilon|z-c'|} = e^{\varepsilon(|z-c'| - |z-c|)} \le e^{\varepsilon \cdot |c-c'|} \le e^{\varepsilon}$, by the triangle inequality. Variance: Var(Lap(b)) = $2b^2 = 2/\varepsilon^2$.

WATCH FOR

- The right chart is the log of the ratio at every possible output. It never leaves the red lines at ±ε.
- Smaller ε widens the noise: more privacy, less accuracy.

Two neighbouring tables

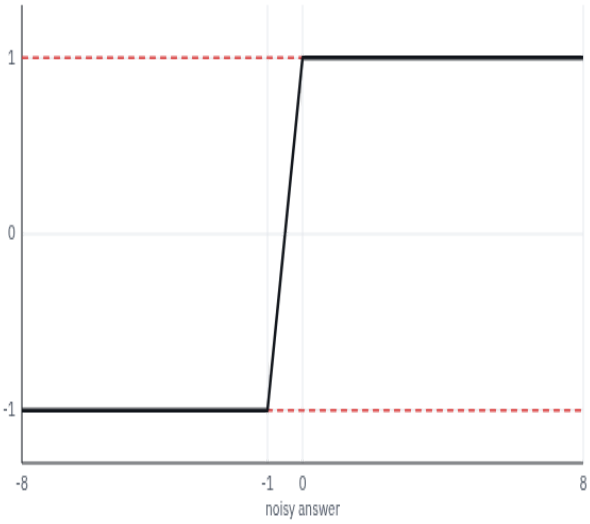
the count differs by 1



output when the table has the person (count c)

output when the person is removed (count c−1)

Privacy loss at each possible output



ln of the density ratio

limits ±ε

NOISE SCALE B = Δ/E

1.00

sensitivity Δ = 1 for a count

STD OF THE NOISE

1.41

Var = $2b^2 = 2/\varepsilon^2$

BIGGEST PRIVACY LOSS

1.000

always ≤ ε = 1.00

WHY IT HOLDS

|z−c| − |z−c'| ≤ 1

triangle inequality gives ratio ≤ e^(ε·1)

Stacking noise: series and parallel

One query is easy. But real use means many queries. How does ε add up?

In series (the same people appear in every query): the budgets **add**, $\varepsilon_{total} = \varepsilon_1 + \varepsilon_2 + \dots$. To keep a fixed total, k queries get ε/k each, so the noise scale is k/ε: it grows linearly with k. The proof multiplies the probabilities of independent answers, so the ratio bounds multiply, and the exponents add.

In parallel (each query on a different group of people): each person is in only one query, so the total is the **maximum**, $\varepsilon_{total} = \max(\varepsilon_1, \varepsilon_2, \dots)$. A histogram over disjoint bins costs ε once, not once per bin.

Fancier analysis (“advanced composition”) lets ε grow like √k instead of k for a small failure probability. I only mention it here.

WATCH FOR

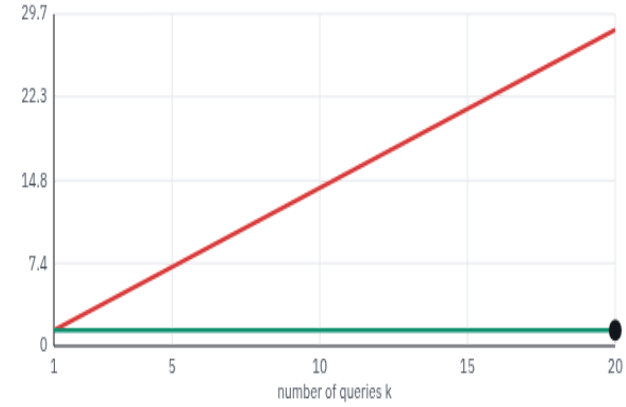
- In series, noise (red) climbs with k; in parallel (green) it stays flat.
- The dots show how many queries touch each person.

Who is touched by the queries?

each person is in exactly 1 of 20 groups



Noise you pay for a fixed total ε



in series (same people)

in parallel (disjoint groups)

your setting

E CHARGED PER QUERY	TOTAL E SPENT	NOISE STD PER ANSWER	COMPARED WITH 1 QUERY
1.000	**1.00**	**1.41**	**1.0×**
each group gets the full budget	$\max(\varepsilon_1,\dots,\varepsilon_k) = 1.000$	Laplace scale $1/\varepsilon = 1.00$, std $= \sqrt{2}/\varepsilon$	no extra noise

The budget against the lecture 1 attacker

Now connect both lectures. A server answers m subset-sum queries with Laplace noise and a total budget ε. By series composition each query gets ε/m, so each answer has noise scale **m/ε**.

The attacker from lecture 1 (random queries + least squares) tries to rebuild the column. If ε is large, the noise is small, so the attacker wins. If ε is small the noise is huge, so the attacker is reduced to guessing (50%).

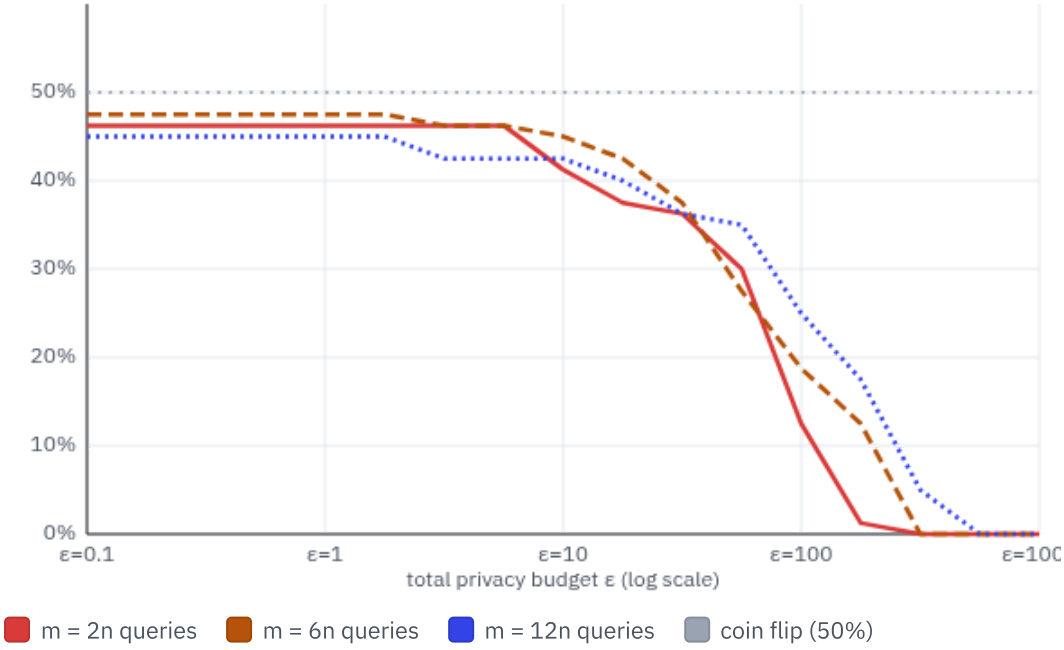
That's the promise of differential privacy, and its price: the more queries you answer, the more total noise you must add.

WATCH FOR

- Each curve starts at about 50% (privacy holds) and falls to 0% (the attacker wins) as ε grows.
- More queries (higher curve at the same ε) means more noise per query.

The lecture 1 attacker against a Laplace server

n = 80, total budget ε split across all queries



What this shows

The server answers m subset-sum queries and splits a total budget ε equally: each gets ε/m, so each answer has Laplace noise of scale m/ε. Large ε means little noise, so the attacker rebuilds the column. Small ε means huge noise, so the attacker is guessing, but then the answers are also useless for the analyst. Privacy and accuracy pull in opposite directions.

Try it: a private query server

A working differentially private server for 200 patients. Each query costs ε from your **budget**; when the budget is spent the server refuses.

- Ask the same question several times: noise is fresh each time and each ask costs budget (series).
- Ask the age-band histogram: four numbers for one ε (parallel).
- The neighbour test on the right plots answers on the real table and on a table with one sick patient removed: the two are nearly indistinguishable.

WATCH FOR

- Lower ε per query: more queries fit in the budget, but each answer is noisier.

Private query server

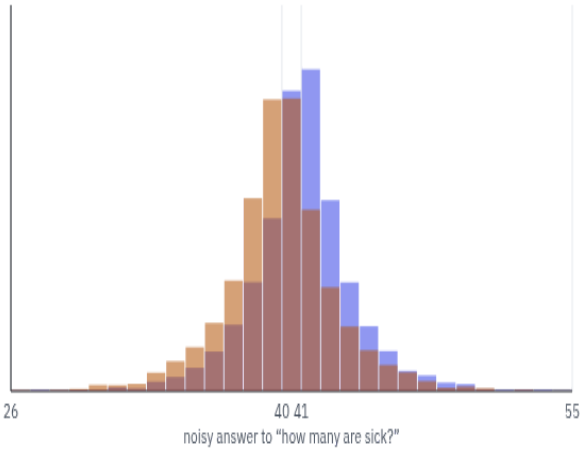
200 patients, secret sick bit

Budget used: 2.00 of 2.00 (0 more queries at this ε)

The 4 bins hold different people, so by **parallel composition** the whole histogram cost only ε = 0.50 even though it returned 4 numbers.

Does one person matter?

the neighbour test



answers over 6000 runs, full table

answers over 6000 runs, one sick patient removed

TRUE COUNTS
41 vs 40
differ by one patient

LARGEST LOG-RATIO SEEN
0.75
limit ε = 0.50 (histogram sampling noise adds a little)

OVERLAP
almost total
can you tell which table produced an answer?

Log

QUERY	TRUE ANSWER (TEACHER VIEW)	NOISY ANSWER	ε CHARGED	ε SPENT SO FAR
Sick by age band (4 bins)	5, 10, 9, 17	6.0, -2.2, 11.0, 9.4	0.50	2.00
Sick count	41	41.2	0.50	1.50
Sick count	41	42.9	0.50	1.00
Sick count	41	39.7	0.50	0.50

What to take away

The cheat sheet on the right has every definition, formula and proof step from this lecture.

Check the details against your lecture notes. If the notes define randomized response or ε-DP slightly differently, tell me and I'll match them.

Back to lecture 1

Cheat sheet

Neighbouring tables D, D′ differ in one person's row.

ε-differential privacy. A randomized mechanism M is ε-DP if for all neighbours D, D′ and every set S of outputs:

$$\Pr[\, M(D) \in S \,] \le e^{\varepsilon} \cdot \Pr[\, M(D') \in S \,]$$

Randomized response (keep the truth with probability $p = e^{\varepsilon}/(1+e^{\varepsilon})$, otherwise flip) is ε-DP: for n people the factors of everyone who didn't change cancel, and the one person who did gives a ratio $\Pr[y|x=0]/\Pr[y|x=1]$ of e^{ε} or $e^{-\varepsilon}$. The full proof is in chapter 5. Two fair coins give $p = 3/4$, $\varepsilon = \ln 3$.

Unbiased estimate from randomized response. $q = \Pr[y=1] = (1-p) + \pi(2p-1)$, so $\hat{\pi} = (\bar{y} - (1-p)) / (2p-1)$ has $E[\hat{\pi}] = \pi$ and $\mathrm{Var}(\hat{\pi}) = q(1-q) / (n(2p-1)^2)$.

Laplace mechanism. For a query f with sensitivity $\Delta = \max |f(D) - f(D')|$, output $f(D) + \mathrm{Lap}(\Delta/\varepsilon)$. Density ratio $\le e^{|f(D)-f(D')|/(\Delta/\varepsilon)} \le e^{\varepsilon}$. Variance $2\Delta^2/\varepsilon^2$.

Composition. Series (same data): $\varepsilon_1 + \varepsilon_2 + \ldots$. Parallel (disjoint data): $\max(\varepsilon_1, \varepsilon_2, \ldots)$. Post-processing never weakens ε.

How lecture 1 and 2 connect

- Lecture 1: if each answer's noise is below about √n, an attacker rebuilds the data.
- Lecture 2: randomized response and Laplace noise give a definition that survives *any* attacker, and composition tells you how fast the budget runs out.
- The variance of a sum of n bits is ∝ n, so its spread is ∝ √n. That is the same √n.

Caveats

I wrote the definitions and proofs from your description of the lecture plus standard references. Constants and exact notation (for example whether the lecture writes e^{ε} or $1+\varepsilon$, or uses the coin version of randomized response) may differ, so compare with the lecture notes. The “advanced composition” result (budget growing like √k instead of k) is beyond what I was told the lecture covered, so it is only mentioned.